

Hospital Chatbot for Color-Coded Clinical Pathways

Using Bidirectional Encoder Representations from Transformers (BERT)

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Outline of Presentation

- 1 Introduction
- 2 Problem, Questions, and Objectives
- 3 Methodology
- 4 Results
- 5 Conclusion
- 6 References



Hospital Overcrowding



Source: GhanaFact

(<https://ghanafact.com/fact-check-korle-bu-teaching-hospital-overwhelmed-with-patients/>)

Problem Statement

Flow management failure

Overcrowding is not only “too many people”: it is a **broken routing system**. Urgency colour does not travel with the patient, and destinations stay unclear after the first desk.

- **No durable colour.** SATS colour assigned at intake often stops guiding beds, labs, and re-assessment timers.
- **Unclear destinations.** Patients and staff lack immediate next-step routing (acute stream vs outpatient vs specialty).
- **Green congestion.** Non-urgent cases occupy acute capacity and delay Red and Orange care.
- **Text before vitals.** Patients arrive with words; TEWS cannot run until measurements exist.



Research Questions

- RQ1** Can the model turn a patient's complaint text into the right urgency level (1–5)?
- RQ2** Can a chatbot give colour and pathway guidance from text before vitals are taken?
- RQ3** Can the system refuse non-medical messages before running the model?



Objectives

- 1 Build a BioBERT classifier that predicts urgency from one English chief complaint.
- 2 Deploy a chatbot that returns colour and pathway from text at intake.
- 3 Add a clinical gate so greetings and non-clinical chat get no colour.



Methodology: NLP Pipeline Overview



Clinical gate

$$g(X) = \text{clip}\left(\sum_j s_j(X) - p(X), 0, 1\right), \quad \text{proceed iff } g(X) \geq 0.35.$$

Rejects non-clinical text (e.g. greetings) before BioBERT runs.

- Softmax yields five acuity probabilities $\hat{y}$ that sum to 1.
- The model picks the urgency level with the highest probability, converts it to a SATS colour (Red, Orange, Yellow, or Green), and attaches a pathway card (where to go and how soon).



Tokenisation, Embeddings, and [CLS]

- WordPiece splits X into subword tokens (max length $M \leq 128$); special tokens [CLS], [SEP], <PAD>.
- Each token gets a 768-d vector by summing word + position + segment embeddings:

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0).$$

- Stack rows $\rightarrow$ input matrix $H^{(0)} \in \mathbb{R}^{M \times 768}$.
- After $L = 12$ encoder layers, classification reads the summary

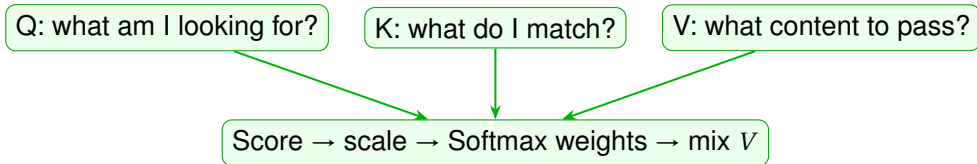
$$h_{[\text{CLS}]} \in \mathbb{R}^{768}.$$



Self-Attention

Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad d_k = 64.$$



Softmax and Colour Map

Five-class head

$$\mathbf{z} = W h_{[\text{CLS}]} + \mathbf{b}, \quad \hat{y}_c = \frac{\exp(z_c)}{\sum_{j=1}^5 \exp(z_j)}.$$

Pick $\hat{c}$ with the largest $\hat{y}_c$; then f_{SATS} : $1 \rightarrow \text{Red}$, $2 \rightarrow \text{Orange}$, $3 \rightarrow \text{Yellow}$, $4, 5 \rightarrow \text{Green}$.

End-to-end

$$X \xrightarrow{g} \tau(X) \rightarrow H^{(0)} \xrightarrow{12} h_{[\text{CLS}]} \rightarrow \hat{\mathbf{y}} \xrightarrow{f_{\text{SATS}}} C_{\text{NLP}} \rightarrow P(C).$$



Model Performance

Holdout evaluation

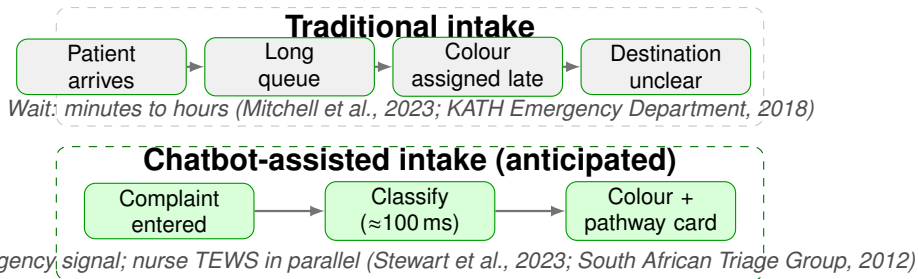
Stratified 10% holdout, $N = 8,000$ English Triagegeist complaints (Laitinen-Fredriksson Foundation, 2025).

Metric (baseline BioBERT)	Value
Accuracy	99.92%
Macro-F1	0.9977
Recall Level 1 (Red)	98.14%
Mean latency	≈ 102 ms / sample

- Strong agreement with nurse acuity labels on held-out English complaints (RQ1).
- Fast enough for prototype intake assistance (RQ2).



Impact on Hospital Intake



- Clearer next steps for urgent vs routine streams; staff remain in control (Sundström et al., 2014; Nurchis et al., 2025).



Conclusion

- A fine-tuned BioBERT model maps English chief complaints to urgency levels 1–5, with 99.92% holdout accuracy and 98.14% Level 1 (Red) recall (Lee et al., 2020; Laitinen-Fredriksson Foundation, 2025).
- The chatbot returns a SATS colour and a pathway card from text in about 100 ms, before vitals are taken; nurses keep TEWS and override (South African Triage Group, 2012; Stewart et al., 2023).
- A clinical gate blocks non-medical messages before the model runs (Stewart et al., 2023).
- The tool supports intake flow; clinicians make the final triage decision (South African Triage Group, 2012).

Limitations. The chatbot only reads English. Test scores come from a held-out practice dataset, not from a live hospital time study. Staff should check unusual wording. Clinicians make the final triage decision.



Selected References

- ① Lee et al. (2020). BioBERT for biomedical text mining. *Bioinformatics*.
- ② Vaswani et al. (2017). Attention is all you need. *NeurIPS*.
- ③ Stewart et al. (2023). NLP at ED triage: a narrative review. *PLOS ONE*.
- ④ South African Triage Group (2012). *SATS Training Manual*.
- ⑤ Chawla et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique.
- ⑥ Laitinen-Fredriksson Foundation (2025). Triagegeist (Kaggle).
- ⑦ Mitchell et al. (2023); Rominski et al. (2014); Berkowitz et al. (2019).
- ⑧ Sundström et al. (2014); Nurchis et al. (2025); KATH SATS implementation reports. 